

CHEATSHEET · QUICK REFERENCE

Lab 3 — LVM Provisioning and Capacity Planning

Build the PV to VG to LV stack, grow XFS and ext4 volumes online, put swap on a dedicated LV and take a point-in-time snapshot.

PV to VG to LV provisioning

```
pvcreate /dev/loop1 /dev/loop2 /dev/loop3
Initialise physical volumes

vgcreate vg_data /dev/loop1 /dev/loop2 /dev/loop3
Create the volume group

lvcreate -L 800M -n lv_web vg_data
Carve a logical volume

lvcreate -L 400M -n lv_db vg_data
Second logical volume

lvs
List logical volumes
```

Format, mount and grow online

```
mkfs.ext4 /dev/vg_data/lv_web
ext4 filesystem

mkfs.xfs /dev/vg_data/lv_db
XFS filesystem

vgextend vg_data /dev/loop4
Add a new PV to the VG

lvextend -L +400M /dev/vg_data/lv_db
Grow the LV

xfs_growfs /srv/db
Grow XFS online

resize2fs /dev/vg_data/lv_web
Grow ext4 online
```

Swap LV and snapshots

```
lvcreate -L 200M -n lv_swap vg_data
Dedicated swap LV

mkswap /dev/vg_data/lv_swap && swapon /dev/vg_data/lv_swap
Enable swap

lvcreate -L 100M -s -n lv_web_snap /dev/vg_data/lv_web
Point-in-time snapshot

mount -o ro /dev/vg_data/lv_web_snap /mnt/snap
Mount snapshot read-only to restore
```

Swap sizing guidance (Red Hat)

RAM	Recommended swap
<= 2 GB	2x RAM
2-8 GB	1x RAM
> 8 GB	0.5x RAM
Databases	Often disabled entirely